The ribbon operator $R_e(t)$ is a multiplication operator exactly at $t = -1$

A Pieri rule for $\langle s_{\mu/\lambda},\ f_e(t)\rangle$, and the origin of the rigid factor $(1+t)$

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Abstract

Let $R_e(t)$ be the operator on the ring Λ of symmetric functions whose matrix in the Schur basis has (μ, λ) entry $t^{h(\mu/\lambda)}$ when μ/λ is an e-ribbon of height h, and 0 otherwise. We compute, in closed form, the pairing of a skew Schur function of size e against $f_e(t) := \sum_{k=0}^{e-1} t^k s_{(e-k,1^k)}$: it is $t^h(1+t)^{c-1}$ when the skew shape has no 2×2 square, c connected components and total height h, and 0 otherwise (Theorem 11). Four consequences. (i) $R_e(-1)$ is multiplication by p_e — this is the Murnaghan–Nakayama rule, and the proof given here is a new one, by an explicit two-element description of the splittings of a ribbon into a horizontal and a vertical strip. (ii) $M_{f_e(t)} - R_e(t) = (1+t)E_e(t)$ with $E_e(t)$ explicit and supported exactly on the disconnected skew shapes. (iii) $[R_e(t), R_{e'}(t)]$ is divisible by $(1+t)$ and, computationally, by nothing more: the rigid factor of the programme is the statement that $t = -1$ is the point at which R_e becomes a multiplication operator, and $\{R_e(-1)\}_e$ commutes because $\{p_e\}_e$ does. (iv) $R_e(t)$ is a multiplication operator *if and only if* $t = -1$; in particular it is not one at the Heisenberg point $t = -q^{-1}$ unless $q = 1$, so the two rigid factors of the programme, $(1+t)$ and $(1+qt)$, are genuinely different phenomena. We also correct an error in the brief that generated this session, and record a prior-art verdict that is deliberately left open.

1 What this session was asked, and what it found

The brief (`state/PROVE.md`, 2026-09-03 cycle 2) posed four targets, T1–T4, around a single observation: the matrix entry of $R_e(t)$ at (μ, λ) depends only on the skew shape μ/λ, which is exactly the condition for an operator to be multiplication by a symmetric function. It asked whether $R_e(t)$ *is* such an operator, predicted that it is not, and predicted that the defect is divisible by $(1+t)$.

All four targets are settled. Two of them are settled in a stronger form than asked; one of them (T4) is settled, but the *inference* the brief attached to it is wrong, and I say so in §7. The mathematical content is Theorem 11, from which everything else falls out in a few lines.

Throughout, Λ is the ring of symmetric functions over $\mathbb{Z}[t]$ with the Hall inner product $\langle \cdot, \cdot \rangle$, for which the Schur functions are orthonormal. For $f \in \Lambda$ we write M_f for the operator "multiply by f".

Definition 1. For $0 \le h \le e-1$ let $N_e^{(h)}$ be the operator on Λ with $N_e^{(h)} s_\lambda = \sum_\mu s_\mu$, the sum over those μ obtained from λ by **adding** an e-ribbon spanning exactly $h+1$ rows. Put $R_e(t) = \sum_{h=0}^{e-1} t^h N_e^{(h)}$.

Here an *e-ribbon* (rim hook, border strip) is a connected skew shape of size e containing no 2×2 square, and its *height* h is one less than its number of rows. Specialising t recovers the operators of the programme: $P_e = R_e(-q)$ and $B_{-1}^{[e]} = R_e(-q^{-1})$.

2 Phase 1: a correction to my own record

The dream cycle of 2026-09-03 recorded the anchor of this family as $R_e(-1) = p_e^\perp$, with the emphatic gloss that $N_e^{(h)}$ *removes* an e-ribbon and that "direction is the whole content". That is backwards. Definition 1 — my own, in `proofs/2026-08-31-Q59-commutator-rigidity.tex` — says *adds*, and the code (`probes/2026-09-03-Q63-form-ii/fock_ell.py::R_e`) moves an abacus bead $b \mapsto b+e$, which adds. The correct anchor is

$$R_e(-1) \;=\; M_{p_e},$$

multiplication, not the adjoint, and it is exact: the Murnaghan–Nakayama sign is $(-1)^{\mathrm{ht}}$ with $\mathrm{ht} = \mathrm{rows} - 1 = h$, so there is no global sign offset to discount. The "$(-1)^h$ versus $(-1)^{h-1}$ discounted kernel" the dream recorded was an artefact of the flipped direction.

Confirmation, with the variation named in advance. Computed entrywise for $e \in \{2,3,4\}$ and $|\lambda| \le 8$:

	$e=2$	$e=3$	$e=4$
$R_e(-1) = M_{p_e}$, entries agreeing	410/410	752/752	1278/1278
NC1: $p_e^\perp$ (the flipped direction) *disagrees*	94/156	69/155	52/137
NC2: $R_e(+1)$ vs p_e *disagrees*	114/410	90/752	160/1278
NC3: perturbed height $h' = \mathrm{rows}$ *disagrees*	228/410	270/752	320/1278

All three controls fire, so the test is not a kernel; NC3 is the one that tests the height grading, which is the part of the object that is mine. The right-hand side was computed twice by routes sharing no code — once through the hook expansion $p_e = \sum_k (-1)^k s_{(e-k,1^k)}$, once by monomial-basis multiplication by $p_e = m_{(e)}$, which never mentions hooks — agreeing on 480/480 entries.

Remark 2. The interesting part of this correction is how it was caught. The dream that recorded the inversion had, that same day, applied the rule "read the raw artifact, not the line about it" to five of *other people's* documents — and inverted its own definition while doing so, in the act of correcting a sub-agent summary that had it right. My own definitions are primary sources, and a correction made while correcting someone else still needs its own warrant. Nothing outside the memory layer was affected: $p_e^\perp$ never entered a paper or a registry node.

3 The candidate is forced, not fitted

Before conjecturing anything, note that the candidate multiplier is determined.

Lemma 3. *If $R_e(t) = M_g$ for some $g \in \Lambda \otimes \mathbb{Z}[t]$, then $g = f_e(t) := \sum_{k=0}^{e-1} t^k\, s_{(e-k,1^k)}$.*

Proof. Take $\lambda = \varnothing$. Then $\langle s_\mu, g\rangle = (M_g)_{\mu\varnothing} = R_e(t)_{\mu\varnothing}$, which is t^h if μ is itself an e-ribbon and 0 otherwise. A partition that is an e-ribbon is exactly a hook $(e-k, 1^k)$, of height k. Since the s_μ are orthonormal, $g = \sum_k t^k s_{(e-k,1^k)}$. □

So the column of $R_e(t)$ over the empty partition writes the candidate down for us; there is no tuning step here, which matters, because a detector that picks its own parameter cannot confirm anything.

Note the specialisations $f_e(-1) = p_e$ (hook expansion of the power sum), $f_e(0) = h_e$, and $[t^{e-1}]f_e = e_e$.

4 A generating function, and what $f_e(t)$ is

Lemma 4. *For $e \geq 1$,*

$$(1+t)\, f_e(t) \;=\; \sum_{b=0}^{e} t^b\, h_{e-b} e_b, \qquad \textit{equivalently} \qquad \sum_{e \geq 0} (1+t) f_e(t)\, z^e \;=\; H(z)E(tz),$$

where $H(z) = \sum h_n z^n$, $E(z) = \sum e_n z^n$ and $f_0(t) := (1+t)^{-1}$.

Proof. Write $S_k = s_{(e-k,1^k)}$. Pieri's rule gives, for $a \geq 1$ and $b \geq 1$, $h_a e_b = s_{(a,1^b)} + s_{(a+1,1^{b-1})}$: indeed $\nu/(1^b)$ is a horizontal a-strip iff $\nu_1 \geq 1 \geq \nu_2 \geq 1 \geq \cdots \geq 1 \geq \nu_{b+1} \geq 0$, which forces $\nu = (a, 1^b)$ or $\nu = (a+1, 1^{b-1})$. Hence $h_{e-b} e_b = S_b + S_{b-1}$ for $1 \leq b \leq e-1$, while $h_e e_0 = S_0$ and $h_0 e_e = e_e = S_{e-1}$. Therefore

$$\sum_{b=0}^{e} t^b h_{e-b} e_b = S_0 + \sum_{b=1}^{e-1} t^b \big(S_b + S_{b-1}\big) + t^e S_{e-1} = f_e(t) + t\big(f_e(t) - t^{e-1} S_{e-1}\big) + t^e S_{e-1} = (1+t) f_e(t). \qquad \square$$

Corollary 5 (identification, gated — see §8)**.** *$f_e(t) = P_{(e)}(x; -t)$, the one-row Hall–Littlewood P-function at parameter $-t$; equivalently $q_e(x;t) = (1-t) f_e(-t)$.*

Proof. Macdonald III (2.10) defines $q_r(x;t)$ by $\sum_{r \geq 0} q_r(x;t) z^r = \prod_i (1 - t x_i z)/(1 - x_i z) = H(z)E(-tz)$, and $q_r = (1-t) P_{(r)}(x;t)$. Substituting $t \mapsto -t$ in Lemma 4 gives $\sum_e (1-t) f_e(-t) z^e = H(z)E(-tz)$. Comparing coefficients, $q_e(x;t) = (1-t) f_e(-t)$; comparing with $q_e = (1-t) P_{(e)}(x;t)$ gives $f_e(-t) = P_{(e)}(x;t)$, i.e. $f_e(t) = P_{(e)}(x;-t)$. $\square$

Two independent specialisations pin this identification rather than merely shape-matching it: $P_{(e)}(x;0) = s_{(e)} = h_e = f_e(0)$ (the degenerate point, which is an anchor but blind), and $P_{(e)}(x;1) = m_{(e)} = p_e = f_e(-1)$ (a second, non-degenerate point). Nevertheless see §8: the two facts quoted from Macdonald III were *not* re-read in this session, and nothing below depends on them — Lemma 4 is self-contained.

5 The main theorem

Fix a skew shape $S = \mu/\lambda$ with $|S| = e$. Write $c(S)$ for its number of connected components (edge-adjacency), and, when S has no 2×2 square, $h(S) = \sum_i (r_i - 1)$ where r_i is the number of rows of the i-th component.

Definition 6. A *splitting* of S is a partition ν with $\lambda \subseteq \nu \subseteq \mu$ such that ν/λ is a horizontal strip and μ/ν is a vertical strip. Put

$$G_S(t) \;=\; \sum_{b \geq 0} t^b \cdot \#\{\text{splittings with } |\mu/\nu| = b\}.$$

Lemma 7. $(1+t)\,\langle s_S,\, f_e(t)\rangle \;=\; G_S(t)$.

Proof. $\langle s_{\mu/\lambda},\, fg\rangle = \sum_\nu \langle s_{\nu/\lambda}, f\rangle\, \langle s_{\mu/\nu}, g\rangle$, since $\Delta s_{\mu/\lambda} = \sum_\nu s_{\nu/\lambda} \otimes s_{\mu/\nu}$ and the Hall pairing makes Λ self-dual with product adjoint to coproduct. By Pieri and dual Pieri, $\langle s_\theta, h_a\rangle$ is 1 if θ is a horizontal a-strip and 0 otherwise, and $\langle s_\theta, e_b\rangle$ is 1 if θ is a vertical b-strip and 0 otherwise. So $\langle s_S, h_{e-b} e_b\rangle$ counts the splittings with $|\mu/\nu| = b$, and Lemma 4 finishes. $\square$

Lemma 8 (a 2×2 kills every splitting)**.** *If S contains a 2×2 square then $G_S(t) = 0$.*

Proof. Suppose the cells $(i,j),(i,j+1),(i+1,j),(i+1,j+1)$ all lie in S, and let ν be a splitting. Row $i+1$ of S is the interval of columns $(\lambda_{i+1},\mu_{i+1}]$, and it contains j and $j+1$, so $\lambda_{i+1}<j$ and $\mu_{i+1}\geq j+1$. The part of μ/ν in row $i+1$ is $(\nu_{i+1},\mu_{i+1}]$; as μ/ν is a vertical strip this has at most one cell, so $\nu_{i+1}\geq\mu_{i+1}-1\geq j$. Hence $(i+1,j)\in\nu/\lambda$. Also $\nu_i\geq\nu_{i+1}\geq j$ and, since $(i,j)\in S$, $\lambda_i<j$; hence $(i,j)\in\nu/\lambda$ too. But ν/λ is a horizontal strip and these two cells lie in the same column j — a contradiction. So there are no splittings. $\square$

Lemma 9 (splittings factor over components)**.** *$G_S(t)=\prod_i G_{S_i}(t)$, the product over the connected components of S.*

Proof. Distinct components of a skew shape occupy pairwise disjoint sets of rows and pairwise disjoint sets of columns. It therefore suffices to check that the interval $\{\nu:\lambda\subseteq\nu\subseteq\mu\}$ is the product of the corresponding intervals, i.e. that the constraint $\nu_i\geq\nu_{i+1}$ never couples two components. Rows with $\lambda_i=\mu_i$ force $\nu_i=\lambda_i$. If rows $i<j$ lie in different components with no constrained row between them, then $j=i+1$ and the two rows share no column, so $\mu_{i+1}\leq\lambda_i$, whence $\nu_i\geq\lambda_i\geq\mu_{i+1}\geq\nu_{i+1}$ automatically; if there is a row k with $i<k<j$ and $\lambda_k=\mu_k$, the same argument applies to the two adjacent pairs. The horizontal- and vertical-strip conditions are conditions on columns and on rows respectively, hence also componentwise. $\square$

Lemma 10 (a ribbon has exactly two splittings)**.** *If S is an e-ribbon of height h then $G_S(t)=t^h(1+t)$.*

Proof. Let the rows of S be $1,\dots,r$ from the top, $r=h+1$, with row i occupying columns $(\lambda_i,\mu_i]$ and $a_i=\mu_i-\lambda_i\geq 1$. Being a ribbon means consecutive rows overlap in exactly one column and there is no 2×2, i.e. $\mu_{i+1}=\lambda_i+1$ for $1\leq i<r$; non-consecutive rows share no column.

A splitting is the same thing as a vector $\alpha=(\alpha_1,\dots,\alpha_r)$ with $\alpha_i=\nu_i-\lambda_i\in[0,a_i]$, the cells of ν/λ in row i being the leftmost α_i cells of that row. Three conditions:

(a) *ν is a partition:* $\lambda_i+\alpha_i\geq\lambda_{i+1}+\alpha_{i+1}$. Using $\lambda_{i+1}=\mu_{i+1}-a_{i+1}=\lambda_i+1-a_{i+1}$ this reads $\alpha_i\geq\alpha_{i+1}-a_{i+1}+1$, i.e. (since $\alpha_{i+1}\leq a_{i+1}$) it is vacuous unless $\alpha_{i+1}=a_{i+1}$, in which case it says $\alpha_i\geq 1$.

(b) *ν/λ is a horizontal strip:* the only column that rows i and $i+1$ share is $\lambda_i+1=\mu_{i+1}$, which lies in ν/λ from row i iff $\alpha_i\geq 1$ and from row $i+1$ iff $\alpha_{i+1}=a_{i+1}$. Combined with (a), which forces $\alpha_i\geq 1$ whenever $\alpha_{i+1}=a_{i+1}$, the horizontal-strip condition is exactly $\alpha_{i+1}\leq a_{i+1}-1$ for $1\leq i<r$; that is, $\alpha_j\leq a_j-1$ for all $j\geq 2$.

(c) *μ/ν is a vertical strip:* $a_i-\alpha_i\leq 1$ for all i.

For $j\geq 2$, (b) and (c) together force $\alpha_j=a_j-1$. For $j=1$ only (c) applies, giving $\alpha_1\in\{a_1-1,a_1\}$; and (a) is then vacuous, since $\alpha_j=a_j$ occurs at most for $j=1$, which has no row above it. So there are exactly two splittings, with $b=|\mu/\nu|=\sum_i(a_i-\alpha_i)=(a_1-\alpha_1)+(r-1)\in\{h,h+1\}$, one of each. $\square$

Theorem 11. *Let $S=\mu/\lambda$ be a skew shape with $|S|=e$. Then*

$$\langle s_S,\ f_e(t)\rangle\ =\ \begin{cases} t^{h(S)}\,(1+t)^{\,c(S)-1}, & \text{if } S \text{ contains no } 2\times 2 \text{ square,}\\ 0, & \text{otherwise.}\end{cases}$$

Proof. By Lemma 7 it suffices to show $G_S(t) = t^h(1+t)^c$ in the first case and 0 in the second. If some component of S contains a 2×2 square, that component's G vanishes by Lemma 8, hence so does G_S by Lemma 9. Otherwise every component is connected with no 2×2, i.e. a ribbon, and Lemmas 9 and 10 give $G_S = \prod_i t^{h_i}(1+t) = t^h(1+t)^c$. $\square$

Equivalently: multiplication by $f_e(t)$ is the operator

$$M_{f_e(t)}\, s_\lambda \;=\; \sum_{\mu} t^{h(\mu/\lambda)}(1+t)^{c(\mu/\lambda)-1}\, s_\mu,$$

the sum over $\mu \supseteq \lambda$ with μ/λ of size e and free of 2×2 squares — i.e. over disjoint unions of ribbons. It is a one-parameter Pieri rule interpolating between h_e (at $t=0$), the Murnaghan–Nakayama rule (at $t=-1$) and e_e (leading coefficient).

6 Consequences

Corollary 12 (T1; Murnaghan–Nakayama)**.** *$R_e(-1) = M_{p_e}$. Indeed $f_e(-1) = p_e$, and at $t=-1$ Theorem 11 gives $(-1)^h 0^{c-1}$, which is $(-1)^h$ when $c=1$ and 0 when $c\geq 2$ or a 2×2 is present. That is precisely the statement $p_e\, s_\lambda = \sum_{\mu/\lambda\ e\text{-ribbon}} (-1)^{\mathrm{ht}(\mu/\lambda)} s_\mu$.*

So the argument above is a proof of the Murnaghan–Nakayama rule, and a rather transparent one: the sign-reversing cancellation that usually carries that proof is replaced by Lemma 10's observation that a ribbon has exactly two splittings, in adjacent degrees, and $(1+t)$ divides out.

Corollary 13 (T2, in the strong form)**.** *$M_{f_e(t)} - R_e(t) = (1+t)\, E_e(t)$, where $E_e(t)$ has (μ,λ) entry $t^{h(\mu/\lambda)}(1+t)^{c(\mu/\lambda)-2}$ when μ/λ has size e, no 2×2 square and $c\geq 2$ components, and 0 otherwise. In particular $E_e(t)$ has polynomial entries and is supported* exactly *on the skew shapes of size e that are not e-ribbons but are disjoint unions of ribbons; on genuine ribbons the defect is not merely divisible by $(1+t)$, it is zero.*

Corollary 14 (the two special points are different)**.** *For $e \geq 2$, $R_e(t)$ is a multiplication operator if and only if $t=-1$.*

Proof. By Lemma 3 the only candidate is $f_e(t)$, and by Corollary 13 the defect vanishes iff $t^h(1+t)^{c-1} = 0$ for every μ/λ of size e with $c \geq 2$ components and no 2×2. Taking $\lambda = (1)$ and $\mu = (e,1,1)$ gives such a shape with $c=2$ and $h=0$ (for $e\geq 2$), whose entry is $(1+t)$; so $t=-1$ is necessary, and by Corollary 12 it is sufficient. $\square$

This separates the programme's two rigid factors. The factor $(1+qt)$ found in Q59 for $[e_i, R_e(t)]$ vanishes at $t=-q^{-1}$, the Heisenberg point $B^{[e]}_{-1}$; the factor $(1+t)$ here vanishes at the Murnaghan–Nakayama point. They coincide only at $q=1$. In particular $R_e(-q^{-1})$ is *not* multiplication by anything for $q\neq 1$, so the $(1+qt)$ of Q59 cannot be explained by the mechanism of this paper, and should not be.

Corollary 15 (T3)**.** *$(1+t)$ divides every entry of $[R_e(t), R_{e'}(t)]$. Explicitly, writing $R_e = M_{f_e} - (1+t)E_e$ and using $[M_{f_e}, M_{f_{e'}}] = 0$,*

$$[R_e(t), R_{e'}(t)] = -(1+t)\Big([M_{f_e}, E_{e'}] + [E_e, M_{f_{e'}}]\Big) + (1+t)^2[E_e, E_{e'}].$$

This is the structural explanation the programme wanted, and it is worth being honest about how cheap it is: divisibility by $(1+t)$ of a polynomial is vanishing at $t=-1$, and at $t=-1$ every R_e is multiplication by a power sum, and power sums commute. The content is not the divisibility; it is Corollary 14, that $t=-1$ is the *only* such point, so this mechanism explains $(1+t)$ and nothing else.

7 T4, and what it does and does not refute

The number. $[R_2(t), R_3(t)]$ is nonzero at generic t. On $|\lambda| \le 10$ it has 531 nonzero entries among 2331 reachable (λ, μ) pairs. The smallest witness is $\lambda = \varnothing$:

$$[R_2(t), R_3(t)]\, s_\varnothing \;=\; (1-t^2)\, s_{(3,2)} \;+\; t(1+t)\, s_{(3,1,1)} \;+\; t(t-1)(t+1)\, s_{(2,2,1)}$$

and one checks the $s_{(3,2)}$ coefficient by hand: $R_2(t)R_3(t)s_\varnothing$ contains $s_{(3,2)}$ (add the row (3), then the domino in row 2) while $R_3(t)R_2(t)s_\varnothing$ does not (a horizontal 3-ribbon cannot be added in row 2 of (2)). Every entry vanishes at $t = -1$ (2331/2331), and the gcd of all nonzero entries is exactly $t+1$, not $(t+1)^2$: the rigid factor is sharp. The case $e = e'$ is the trivial kernel and gives 0, as it must.

What this does not show. The brief asserted that a nonzero commutator refutes Q69, because Leclerc–Thibon's operators satisfy $[V_i, V_j] = 0$. That inference is wrong, and I checked it against the primary source rather than against my note about it. Thibon's LeclercFest slides (`sources/thibon/blfest.pdf`, read this session) define, for a fixed ribbon size r,

$$V_k = \text{“multiplication by } p_r \circ h_k\text{”}, \qquad V_k\, s_\lambda = \sum (-1)^{h(\mu/\lambda)} s_\mu$$

summed over μ with μ/λ a *horizontal r-ribbon strip of weight k*, where $h(\mu/\lambda) = \sum_R (h(R) - 1)$ over the r-ribbons R tiling it. So the family $\{V_k\}_k$ is indexed by the *weight* k at fixed r, while $[R_e, R_{e'}]$ varies the *ribbon size*. These are different indices; $[V_i, V_j] = 0$ says nothing about $[R_e, R_{e'}]$, and a witness on the wrong axis is not a witness.

What is true, which is better. At $k = 1$ a horizontal r-ribbon strip of weight 1 is a single r-ribbon and $h = \text{rows} - 1$, so the slides give

$$V_1 = R_r(-1) = M_{p_r} \quad (\text{consistent with } p_r \circ h_1 = p_r), \qquad V_1^{(q)}|\lambda\rangle = \sum (-q)^{-h}|\mu\rangle = R_r(-q^{-1})|\lambda\rangle = B_{-1}^{[r]}|\lambda\rangle.$$

So Q69's identification holds, at the single point $t = -q^{-1}$, and it is prior art (Leclerc–Thibon), not new. The right statement is a *dissolution* rather than an answer: $\{V_k\}_k$ and $\{R_e(t)\}_t$ are two one-parameter deformations of the same operator M_{p_e} in different directions. The V_k direction deforms the plethysm index and stays inside the multiplication operators, which is why it commutes; the t direction deforms the height grading and leaves the multiplication operators immediately, by Corollary 14. The two families meet at exactly one point. Q72's winding-number separator is therefore not needed for this purpose.

8 Prior art: an open verdict, deliberately

Theorem 11 is, via Corollary 5, a Pieri rule for the one-row Hall–Littlewood function against skew Schur functions: $\langle s_{\mu/\lambda}, q_e(x;t)\rangle = (-t)^h(1-t)^c$ for μ/λ free of 2×2 squares, and 0 otherwise. This is elementary enough that I expect it to be classical, and I am not claiming it as new.

I could not settle this in a proof session (no browsing). Concretely, what is *not* established here:

1. The two facts quoted from Macdonald III in Corollary 5 — the generating function $\sum_r q_r(x;t)z^r = \prod_i (1 - tx_iz)/(1 - x_iz)$ and $q_r = (1-t)P_{(r)}$ — were cited from memory; Macdonald is not on disk in this container. *Nothing else in this paper depends on them*: Lemma 4 proves the generating function for $f_e(t)$ from Pieri, and Theorem 11 is stated and proved in terms of $f_e(t)$ alone. Corollary 5 is a remark, and should be re-checked against Macdonald III.2 before being used.

2. Whether Theorem 11 itself appears in the literature. Search terms for the next browse: a Murnaghan–Nakayama or Pieri rule for Hall–Littlewood $q_r/P_{(r)}$ expanded against *skew* Schur functions; the $(1-t)^c$ weight on disconnected border strips; Morris' recursion. Local sources hold nothing: the only Hall–Littlewood material on disk is Thibon's slides, whose HL sections concern $\tilde{K}_{\lambda\mu}(q)$, cospin and unipotent varieties, not this pairing.

The honest verdict is therefore **unsettled, flagged** — not "new". The value of the result to the programme is unaffected either way: what is being used downstream is Corollary 14, which is a statement about $R_e(t)$.

9 Computational verification

All computations use `probes/2026-09-03-Q75/`, built on a small symmetric-function engine (`symfunc.py`) deliberately independent of both the Murnaghan–Nakayama rule and the abacus code, since those are what is being tested: it is built from Kostka numbers by direct enumeration of horizontal strips, the monomial basis, and dominance-triangular Schur decomposition.

Claim	Range	Result
Lemma 4, $(1+t)f_e = \sum_b t^b h_{e-b} e_b$	$e \le 5$	exact
$G_S(t) = t^h(1+t)^c$ resp. 0 (Lemmas 8–10)	$e \in \{2,3,4\}$, $\lvert\lambda\rvert \le 7$	1482/1482
Theorem 11, entrywise	$e \in \{2,3,4\}$, $\lvert\lambda\rvert \le 8$	2440/2440
Corollary 12, $R_e(-1) = M_{p_e}$	$e \in \{2,3,4\}$, $\lvert\lambda\rvert \le 8$	2440/2440
Corollary 13, $(1+t) \mid (M_{f_e} - R_e)$	same	2440/2440, 1567 nonzero
$[R_e, R_{e'}]$ vanishes at $t=-1$	$(2,3)$, $\lvert\lambda\rvert \le 10$	2331/2331
$[R_e, R_{e'}] \neq 0$	$(2,3)$, $\lvert\lambda\rvert \le 10$	531 nonzero entries
gcd of entries of $[R_e, R_{e'}]$	$(2,3), (2,4), (3,4)$, $\lvert\lambda\rvert \le 8$	$t+1$ exactly

The separator, run before anything was believed. The statement $M_{f_e} = R_e + (1+t)E_e$ would be vacuous if the two sides agreed identically, so the variation was named in advance: at $t=0$, $R_e(0)$ adds only a *connected* horizontal e-ribbon, whereas M_{h_e} adds *any* horizontal e-strip. They differ on 62/156 entries at $e=2$, 89/274 at $e=3$, 101/453 at $e=4$ (all with $|\lambda| \le 6$). The separator is live; the theorem is not a statement about the zero operator. The 1567 nonzero defect entries in the table above are the same fact counted differently.

Negative controls. Section 2 lists three for the anchor, all firing. For T4 the trivial kernel $e = e'$ was excluded by hand and checked to give 0 (608/608 entries), confirming that the nonzero commutator is not an artefact of the computation.

10 Gaps

1. Corollary 5 rests on two facts cited from memory (§8, item 1). Everything else is self-contained.

2. The prior-art status of Theorem 11 is unsettled (§8, item 2).

3. The structure of $[R_e(t), R_{e'}(t)]$ beyond divisibility by $(1+t)$ is not determined here. Corollary 15 expresses it in terms of E_e, and the computations say the factor is exactly $(1+t)$, but no closed form for the quotient is proved. This is the natural next target, and it is now a question about E_e, which Corollary 13 makes fully explicit.

4. Nothing here touches the q-deformed factor $(1+qt)$ of Q59. Corollary 14 shows it cannot come from this mechanism.